Just the initial plan. Most of these done by A

COSC2671 Assignment 2 — Work Split

AGP 2025 YouTube Social Network Analysis · Due 22 May 2026 · Team: S, M, A

S — Sona

NLP lead + data pipeline + network construction

CARRIES OVER FROM A1

- ✓ Fetch pipeline (fetchYoutubeData.py, preprocess.py)
- ✓ VADER sentiment analysis + 4 visualisations
- ✓ LDA topic modelling (5 topics) + word clouds

NEW WORK FOR A2

- + build_network.py — bipartite user-video graph → projected user-user network
- + cross_analysis.py — merge network position with sentiment & topic labels
- + Extend sentiment: cross-topic analysis (does topic predict sentiment?)
- + Temporal sentiment: how did sentiment shift around race weekend?

REPORT SECTIONS

- ▶ Introduction, Data collection & preprocessing
- ▶ NLP analysis (sentiment + topics)
- ▶ Cross-analysis findings (network position vs sentiment)

M

Team management lead + information diffusion

NEW WORK FOR A2

- + Set up and maintain GitHub repo — folder structure, README, branches
- + Maintain weekly timesheets for all 3 members throughout project
- + Write project plan + milestone tracker (management/project_plan.md)
- + Coordinate Access .txt file submission for lecturer
- + diffusion.py — Independent Cascade Model on user-user network (L10)
- + Seed diffusion with high-PageRank users — simulate sentiment spread

REPORT / PRESENTATION

- ▶ Discussion, Conclusion, Limitations, References
- ▶ Compile full worksheet (timesheets + self-reflections for all 3)
- ▶ Coordinate presentation — ensure all 3 members speak

A

Network analysis lead + community detection

NEW WORK FOR A2

- + network_analysis.py — degree, betweenness, PageRank, eigenvector centrality (L8)
- + Interpret centrality: who are the most influential commenters?
- + Clustering coefficients — how tightly clustered is the fan network?
- + network_viz.py — NetworkX visualisations, node size = PageRank
- + community_detection.py — Louvain modularity, label per user (L9)
- + Characterise communities — size, density, dominant topics per community

REPORT / PRESENTATION

- ▶ Network construction, Centrality analysis, Community detection sections
- ▶ Lead presentation structure and slide design

SHARED — ALL 3 MEMBERS

◆ Problem formulation ◆ Peer code review ◆ Individual self-reflection ◆ Presentation delivery (each must speak) ◆ Final report proofreading

GITHUB REPO STRUCTURE

```
AGP2025-SMNA-A2/
  code/
    [A1 scripts]                ← S       network_analysis.py         ← A
    build_network.py            ← S       community_detection.py      ← A
    cross_analysis.py           ← S       network_viz.py              ← A
    diffusion.py                ← M
  data/                         (sample only — no full datasets in repo)
  management/                   ← M  (project_plan.md, timesheets.xlsx)
  README.md   requirements.txt
```

Note: Timesheets must be filled in weekly and look realistic — the rubric explicitly flags "manufactured" timesheets as Poor/Fair. M owns this responsibility throughout the project.